

Somantha Manuranga

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PROFESSIONAL SUMMARY

Master's student in AI for Industrial Applications with a background in Mechanical Engineering. Focused on Computer Vision, Model Robustness, and Digital Twins for Industry 4.0. Skilled in Python, PyTorch, database systems, and building production-ready AI pipelines. Seeking a Werkstudent position in AI/ML engineering in Germany.

EDUCATION

OTH Amberg-Weiden University

Master of Artificial Intelligence for Industrial Applications

Germany

2025 - 2027(expected)

- Modules: Computer Vision, Deep Learning, Autonomous Robotics, NLP
- Focus on applying computer vision, deep learning, and machine learning to industrial processes.

University of Sri Jayewardenepura

Bachelor of Science in Mechanical Engineering (Minor in Mechatronics)

Sri Lanka

2019 - 2023

- **Thesis:** Developed a machine learning model for 2D aerodynamic drag prediction in NACA airfoils
- **Project:** CoreXY 3D printer development integrating mechanical design and machine learning models.

PROJECTS

Quantum Circuit Image Dataset & Classification Pipeline | *BeautifulSoup, arXiv API, BERT, NLP* 2026

- Architected an automated end-to-end NLP pipeline, extracting and structuring metadata from 250+ arXiv research papers into JSON.
- Fine-tuned a SciBERT foundation model (Hugging Face) for caption classification, achieving an F1-score of 0.96.
- Enforced data quality and reproducibility through automated validation checkpoints and strict Git version control.

Machine Learning Challenge | *Edge AI, Ensemble Methods, Python, Rule Based System* 2026

- Trained and evaluated ensemble models (XGBoost, LightGBM, CatBoost) for high-dimensional regression, achieving $R^2 = 0.97$.
- Engineered a lightweight, 3-rule interpretable model optimized for edge deployment, maintaining accuracy ($R^2 = 0.98$) under strict resource constraints.

ALIA Delivery - Hospital Digital Twin | *Industry 4.0, Python, MQTT, Webots, Dijkstra's algorithm* 2025

- Designed and implemented a low-latency MQTT communication layer linking a Python web application with a Webots robotics simulation.
- Deployed a real-time digital twin interface featuring autonomous robot navigation and dynamic decision-making capabilities.

CNN-based Drag Prediction for 2D Geometries | *TensorFlow, Python, Neural Network, Deep Learning* 2023

- Developed Convolutional Neural Networks to predict drag coefficients of arbitrary 2D shapes, replacing computationally expensive CFD simulations.
- Trained on 12,000+ laminar-flow samples, achieving a minimum validation loss of 0.0017 and <1% relative error for NACA airfoils.

Industrial Production Analytics Dashboard | *Power BI, DAX, Power Query, SQL, Star Schema* 2025

- Built interactive Power BI dashboards using Star Schema models and automated ETL pipelines to track large-scale manufacturing KPIs for cross-functional teams.

Student Behaviour Analytics - Multivariate EDA | *Python, Matplotlib, Seaborn* 2025

- Conducted multivariate EDA on 700+ records to identify correlations between screen time and academic performance, delivering structured findings to non-technical stakeholders.

EXPERIENCE

KNSI Aviation Engineering

Sri Lanka

Design Engineer

2023 - 2024

- Used SQL and Excel to manage compliance data workflows and technical documentation.
- Developed automated SQL/Excel templates, reducing weekly reporting time by 30%
- Analyzed production data to optimize workflows, resulting in a 40% increase in process efficiency.

Ceylon Biscuits Limited

Sri Lanka

Internship - Mechatronic Engineer

2022 - 2023

- Contributed to process optimization through data-driven solutions and automation projects.
- Collaborated with cross-functional technical teams to troubleshoot operational bottlenecks and enhance daily production output.
- Designed and rapidly prototyped 3D-printed automation components, reducing external part sourcing costs and lead times.

TECHNICAL & LANGUAGE SKILLS

Programming & DevOps: Python (Advanced), C++, SQL, Git/GitHub, REST APIs, pytest
AI, ML & Generative AI: PyTorch, TensorFlow, Scikit-learn, CNN, RNN, Hugging Face, LLMs (Llama, GPT), RAG, LangChain, Agentic Workflows, Fine-tuning
Computer Vision: OpenCV, Image Classification, Object Detection, Feature Extraction
Data & Backend: Pandas, NumPy, MongoDB (NoSQL), MQTT, ETL Pipelines, Power BI
Languages: English (C1 - Professional), German (B1 - Improving), Sinhala (Native)

LEADERSHIP & EXTRACURRICULARS

Football Club Amberg: Active player in a local German football club, FC Amberg
University Colors Award: Awarded University Colors for Sports Excellence in Football (2022) , Sri Lanka
Team Captain: Led Engineering Faculty & University Football Team as Captain, Sri Lanka